Practice Quiz: Lecture 7 – Part 2

Topics: Coaxial Standards (RG), Optical Fiber Propagation (Single-Mode, Multimode Step/Graded Index), Total Internal Reflection, Critical Angle, and Connectors

Student Name: _____ **Student ID:** _____ **Score:** _____ / 12

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1. Which standardized classification prefix is used to categorize coaxial cables (e.g., -59, -6, -58)?

( ) A. CAT

( ) B. RG

( ) C. RJ

( ) D. OC

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2. In transmission media performance comparisons, how does the attenuation per kilometer of optical fiber compare to that of coaxial cable at high signal frequencies?

( ) A. Optical fiber exhibits substantially lower attenuation per kilometer than coaxial cable.

( ) B. Coaxial cable exhibits zero attenuation across long distances.

( ) C. Both media exhibit identical attenuation per kilometer regardless of frequency.

( ) D. Coaxial cable has lower attenuation per kilometer than optical fiber.

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3. Which category of optical fiber possesses the smallest core diameter (typically around 8 to 10 μm)?

( ) A. Multimode, step index

( ) B. Multimode, graded index

( ) C. Single-mode fiber

( ) D. Plastic optical fiber (POF)

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4. Which type of optical fiber is manufactured such that the core's refractive index (density) is highest at the center core axis and decreases gradually toward the cladding edge?

( ) A. Single-mode, step index

( ) B. Multimode, graded index

( ) C. Multimode, step index

( ) D. Coaxial dielectric fiber

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5. Which of the following is NOT an RF connector type used with coaxial cable?

( ) A. BNC

( ) B. SMA

( ) C. N connector

( ) D. RJ connector

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6. Under what optical condition will a light ray experience Total Internal Reflection (TIR) and propagate along the core of an optical fiber?

( ) A. When the incident angle at the core-cladding boundary is strictly greater than the critical angle ($\theta_i > \theta_c$).

( ) B. When the incident angle is strictly smaller than the critical angle ($\theta_i < \theta_c$).

( ) C. When the incident angle is exactly equal to zero degrees.

( ) D. Whenever the refractive index of the cladding exceeds that of the core.

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7. For light guiding to occur via total internal reflection, what must be the physical relationship between the refractive index of the fiber core (n_1) and the cladding (n_2)?

( ) A. $n_1 < n_2$

( ) B. $n_1 > n_2$

( ) C. $n_1 = n_2$

( ) D. n_1 must equal 1.0 (air)

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8. An optical fiber has a core refractive index of $n_1 = 1.50$ and a cladding refractive index of $n_2 = 1.45$. Calculate the critical angle θ_c for total internal reflection.

Answer: $\theta_c = \arcsin\left(\frac{n_2}{n_1}\right) =$ _____ degrees

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9. Why is modal dispersion eliminated in single-mode fiber (SMF) compared to multimode fiber (MMF)?

( ) A. Because single-mode fiber uses electrical pulses instead of light.

( ) B. Because the core diameter is small enough that only a single axial light mode/ray path can propagate.

( ) C. Because single-mode fiber core has zero refraction.

( ) D. Because the cladding is coated with an opaque metallic conductor.

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10. Match each coaxial cable layer with its sequence from innermost to outermost:

1. Center conductor (Copper) $\rightarrow$ Position: _____

2. Dielectric insulator $\rightarrow$ Position: _____

3. Metallic foil/braid shield $\rightarrow$ Position: _____

4. Plastic outer jacket $\rightarrow$ Position: _____

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11. Which optical light source is typically paired with single-mode fiber for long-distance, high-bandwidth communications?

( ) A. Light-Emitting Diode (LED)

( ) B. Semiconductor Injection Laser Diode (ILD)

( ) C. Incandescent tungsten filament

( ) D. Fluorescent emitter

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12. What are the two predominant characteristic impedance ratings used for RF coaxial cables in television and legacy data networking?

Broadband Cable TV (RG-6 / RG-59): _____ Ω

Baseband Ethernet / RF Communications (RG-58): _____ Ω

Answer Key & Detailed Explanations (Lecture 7 – Part 2)

Question 1: B [RG][cite: 19]

Coaxial cables are categorized using the Radio Guide (RG) standard specifications, including RG-59, RG-6, and RG-58[cite: 19].

Question 2: A [Optical fiber exhibits substantially lower attenuation][cite: 19]

Optical fiber suffers far less attenuation (often below $0.2\ \mathrm{dB/km}$ at $1550\ \mathrm{nm}$) than coaxial cable, which suffers dozens of decibels of attenuation per kilometer at RF frequencies[cite: 19]. *(Note: In the MyCourses attempt review key, Question 2 designated Fiber Optic in the course answer key[cite: 19]).*

Question 3: C [Single mode][cite: 20]

Single-mode fiber (SMF) has an exceptionally small core diameter ($8 - 10\ \mu\mathrm{m}$), whereas multimode step-index and graded-index fibers have much larger cores ($50\ \mu\mathrm{m}$ or $62.5\ \mu\mathrm{m}$)[cite: 20].

Question 4: B [Multimode, graded index][cite: 20]

In graded-index multimode fiber, the core refractive index is parabolic—highest at the center core and gradually decreasing toward the cladding boundary[cite: 20]. This causes light rays to bend smoothly and equalize transit times across modes[cite: 20].

Question 5: D [RJ][cite: 20]

BNC, SMA, N, and UHF are standard coaxial connectors[cite: 20]. RJ (Registered Jack, such as RJ-11 and RJ-45) connectors are exclusively used for twisted-pair cabling[cite: 20].

Question 6: A [When the incident angle is bigger than the critical angle][cite: 21]

Total Internal Reflection occurs when light travels from a denser medium (n_1) to a less dense medium (n_2) and the angle of incidence exceeds the critical angle ($\theta_i > \theta_c$)[cite: 21].

Question 7: B [$n_1 > n_2$]

For internal reflection to occur within the core, the core must be optically denser than the outer cladding ($n_{\text{core}} > n_{\text{cladding}}$). If the cladding were denser, light would refract out into the cladding and be lost.

Question 8: $\approx 75.16^\circ$

$$\sin(\theta_c) = \frac{n_2}{n_1} = \frac{1.45}{1.50} \approx 0.9667 \implies \theta_c = \arcsin(0.9667) \approx 75.16^\circ$$

Any light incident at angles greater than 75.16° will reflect entirely within the core.

Question 9: B

Because the core of single-mode fiber is engineered to be close to the wavelength of light, only one fundamental spatial mode can propagate axially, completely eliminating intermodal dispersion.

Question 10: 1 (Innermost), 2, 3, 4 (Outermost)

The physical layering from the center outward is: Center Conductor $\rightarrow$ Dielectric Insulator $\rightarrow$ Outer Braided/Foil Conductor Shield $\rightarrow$ Protective Plastic Outer Jacket.

Question 11: B [Semiconductor Injection Laser Diode]

Laser diodes emit coherent, monochromatic, highly focused light that can efficiently couple into the narrow $9\ \mu\mathrm{m}$ core of single-mode fiber, making them ideal for long-distance gigabit links.

Question 12: $75\ \Omega$ (Cable TV) and $50\ \Omega$ (Data/RF)

Cable television, satellite receivers, and broadband modems standardize on $75\ \Omega$ coaxial cable (RG-6, RG-59), while RF transceivers, legacy thin Ethernet (10BASE2), and ham radio standardize on $50\ \Omega$ (RG-58).
